
Generalization-Memorization Phase Transition for Flow Matching Models

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Abstract

In this paper, we investigate the transition from generalization to memorization in flow matching models. Our results show that flow matching exhibits similar learning dynamics to diffusion models, where models show signs of generalization before they start to memorize training data. The onset of memorization scales proportionally to the number of training data points, with larger datasets delaying the memorization regime. Moreover, the width of flow matching networks i.e., model capacity, is inversely related to the onset of memorization, confirming that the implicit regularizations of diffusion model training apply as well to flow matching models. Furthermore, we find that flow matching reaches a limitation when applied to heavy-tailed data, modeled by the Cauchy distribution, or Student’s t -distribution with degrees of freedom $d_f = 1.0$. Flow matching struggles to generalize and memorizes much of the training data. To counteract the impact to performance under heavy-tailed data, we apply a method to intentionally learn curved trajectories by modifying the time schedule in the loss function calculation. Our results show that imposing a sinusoidal schedule improves generalization on the Cauchy distribution and reduces the degree of memorization on heavy-tailed distributions with $d_f \leq 3.0$. Our results align with prior literature on the importance of understanding the geometry of data manifolds to run more realistic, and by extension accurate, flow matching simulations.

1 Introduction

Flow matching is a frequently used alternative to diffusion for generative modeling. Flow matching provides simpler training objectives and a more efficient sampling method using ODE solvers [19], [26]. However, a foundational understanding on the generalization behavior of these models is largely unknown. There have been many studies published in the last few years on the generalization-memorization transition in diffusion models [3] and on the implicit bias and regularization in diffusion training dynamics [1], [8]. Recent work has identified dataset size and training time as key factors that influence when diffusion models begin to memorize training data after first generalizing [3]. However, similar analyses in the literature focused on flow matching as distinct from diffusion are missing. While the prior literature has linked flow matching and diffusion as equivalent methods of learning generative models, flow matching is formulated and implemented in ways that are fundamentally different, e.g., conditional loss, optimal transport velocities, no noising or denoising, no time reversal, faster inference with ODE solvers [19], [1], [21], [16] [20]. Therefore, we conjecture that implicit biases in flow matching training need not be one and the same with diffusion [8].

2 Flow Matching

Using flow matching, we aim to learn a target distribution π by simulating an ordinary differential equation (ODE). The solution to the ODE is a trajectory $(X_t)_{0 \leq t \leq 1}$ that maps time t to spatial

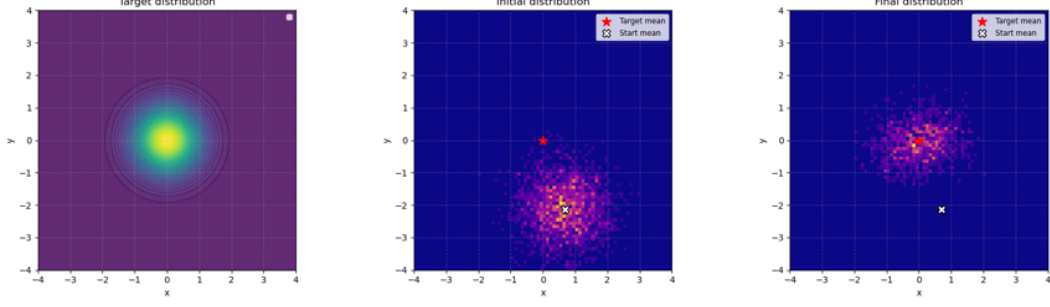


Figure 1: Target Gaussian distribution (left), denoted π . Samples from initial distribution ρ_0 (middle). Samples from the final distribution ρ_1 as a function of ρ_0 and velocity fields u_t^θ parameterized by a neural network (right).

coordinates in the domain, here, $\mathbb{R}^n$ (See Figure 1 for an example) [15]. Each spatial coordinate along the trajectory is drawn from an intermediate distribution $X_t \sim \rho_t$ where the initial conditions are $X_0 \sim \rho_0$ and we obtain $X_1 \sim \rho_1$. The continuity equation for the probability flow is:

$$\frac{\partial}{\partial t} \rho_t = -\nabla(\rho_t u_t)$$

where $u_t : \mathbb{R}^n \times [0, 1] \rightarrow \mathbb{R}^n$ is the velocity field at each timestep t that defines a velocity vector at each location $x \in \mathbb{R}^n$ [15]. The velocity field u_t defines an ODE that constrains the trajectory $(X_t)_{0 \leq t \leq 1}$, and this trajectory is the solution to the ODE, defined as:

$$\frac{d}{dt} X_t = u_t(X_t) dt$$

where $X_0 \sim \rho_0$ at time $t = 0$. While the ODE is deterministic, randomness is introduced by choosing ρ_0 to be a probability distribution, such as the standard Gaussian $\mathcal{N}(0, I_n)$.

While the velocity field describes instantaneous change for each timestep, the trajectory itself is defined by the flow ψ , the mapping from sample X_0 to X_1 , i.e., $\psi : \mathbb{R}^n \times [0, 1] \rightarrow \mathbb{R}^n$ such that:

$$X_t = \psi_t(X_0), t \in [0, 1].$$

The flow can be defined by the velocity field u_t via the ODE:

$$\frac{d}{dt} \psi_t(x) = u_t(\psi_t(x))$$

where $\psi_0(x) = x$ [20]. In summary, the solution to the ODE defined by vector fields is the flow that transports initial samples to the target distribution. Neural networks parameterize the velocity field u_t^θ and an ODE solver is used to find the corresponding flow for all $t \in [0, 1]$.

3 Empirical Results

Experimental Setup Experiments are run on a toy spiral dataset in two dimensions. We use a fully-connected network with a single hidden layer and ReLU activations. The model takes 2-dimensional spatial positions and time as inputs [1]. The forward pass uses a linear projection to encode the time dimension before running spatial and embedded time through the neural network [19]. We use the `torchdiffeq` implementation of the DOPRI5 method to generate new samples [19], [5]. The flow matching architecture stays consistent throughout the study, with the exception of one experiment that varies the hidden layer width. We use conditional loss to train the model using a random interpolant between a training datapoint x_0 and a random point in the domain [15]:

$$\mathcal{L}_{CFM} = \mathbb{E}_{t, \pi(x_0), \rho_t(x_t|x_0)} [\|u_t^\theta(x_t) - u_t^\theta(x_t|x_0)\|^2],$$

where $x_t := (1 - t)x_0 + t\epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, $t \sim \text{Uniform}(0, 1)$. We encourage the model to learn straight velocity paths by using $x_0 - \epsilon$ as the target velocity $u_t^\theta(x_t|x_0)$ to regress against.

Evaluation Metrics Generalization is measured using the Wasserstein distance [7] between the empirical distribution of the training data $\{x_1, \dots, x_n\}$, $x_i \sim \pi$ and that of generated samples $\{y_1, \dots, y_m\}$, $y_j \sim \rho_t$, with the simplification of giving uniform probability weight to each datapoint to avoid solving a linear program [32]:

$$W_1(\rho_t, \pi) = \sum_{i=1}^n \sum_{j=1}^m \|x_i - y_j\|_2$$

The Wasserstein distance is preferred over pointwise divergences such as ϕ -divergences to allow for geometric intuitions on the closeness of empirical distributions without overlapping supports [31].

To align with previous studies on diffusion model learning theory [3], [37], [13], memorization is measured as the fraction of generated data that closely matches its training data. Concretely, a generated point x_t is considered memorized if the following condition is met:

$$\mathbb{E}_{x_t} \left[\frac{\|x_t - a_1\|_2}{\|x_t - a_2\|_2} \right] < 0.50$$

where a_1 and a_2 are the nearest and second nearest neighbors in the training dataset, respectively.

3.1 Generalization-Memorization Phase Transition

Our results show that a similar trend arises for flow matching models as it does for diffusion, where models first show signs of generalization, before memorization sharply rises [3]. For varying levels of training data provided, an inverse relationship between training data size and the onset of memorization is observed (See Figure 2). This indicates that, as in diffusion models, larger amounts of training data delay the memorization threshold [3]. Further, in the memorization regime (epoch $10^3 - 10^5$), we note that the training loss is minimized by memorizing data points, hindering the ability to learn the true manifold in later epochs. We also note a tradeoff between sample accuracy, i.e., the Wasserstein distance with the target distribution, and sample quality, i.e., the fraction of memorized data f_{mem} .

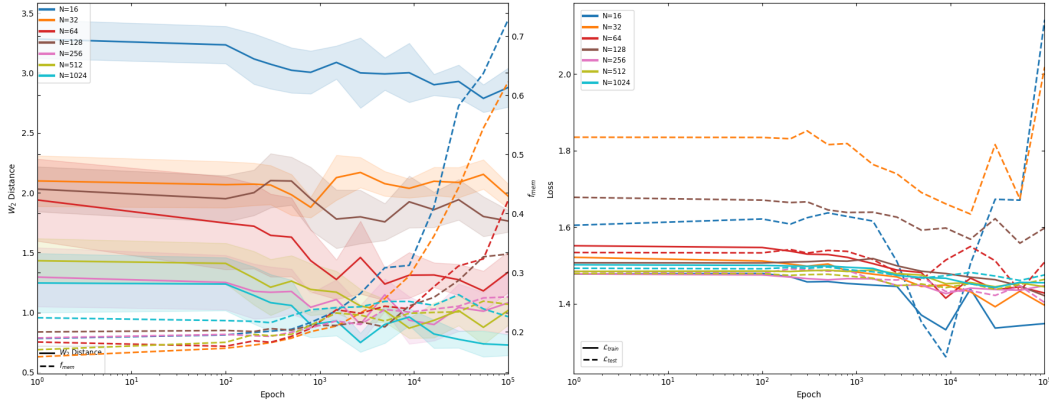


Figure 2: Generalization-Memorization phase transition under varying training datapoints $N \in \{16, 32, 64, 128, 256, 512, 1024\}$, following the approach in [3]. Generalization is measured using Wasserstein distance between generated samples and the target distribution ± 1 standard deviation. Memorization is measured using the fraction of training data that is memorized, denoted f_{mem} (left). We observe that generalization begins even in early training epochs while memorization arises in epochs $10^2 - 10^3$. Train and test loss are plotted to present another view of the onset of memorization as the point the train and test losses diverge (right).

However, for varying levels of model width, we observe that larger model capacity indeed delays the memorization threshold, however, once the threshold is met, the degree of memorization scales proportionally to model width (See Figure 3). This indicates that overparameterized networks contain inductive biases that indeed delay memorization, however once the model starts memorizing, the large capacity of the model allows it to match individual datapoints exactly, allowing larger networks to memorize more than smaller ones. This is in contrast to the literature on diffusion models which

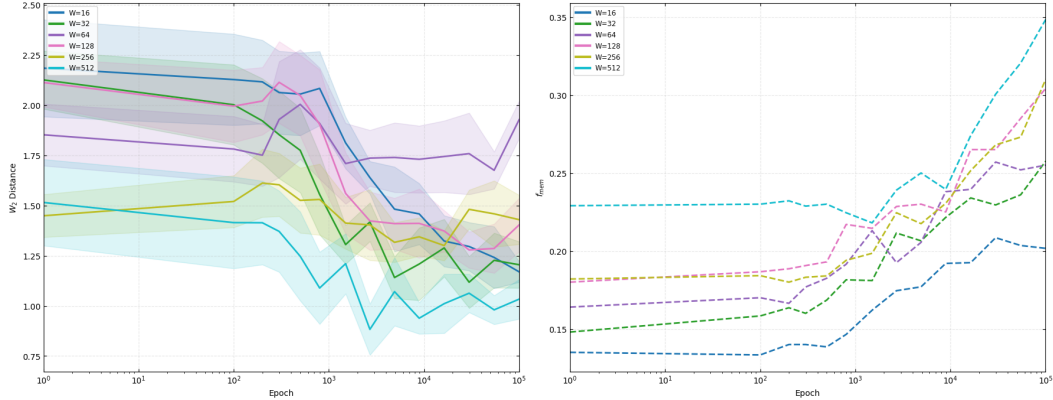


Figure 3: Phase transition under varying model width W in the hidden layer. Values $W \in \{16, 32, 64, 128, 256, 512\}$ are shown. Generalization begins in early epochs (left), and memorization arises in epoch $10^2 - 10^3$ (right).

does not explore the degree of memorization once the threshold has been met. This opens the question of whether the deterministic nature of flow matching limits the ability of models to handle noise, outliers, or other artifacts in the training data, and whether diffusion models can better handle outliers data due to the stochasticity of their trajectories.

3.2 Effect of Heavy-Tailed Data on Training Dynamics

From our own results and as suggested by [3], the generalization-memorization phase transition is not well studied under assumptions of heavy-tailedness. To explore how flow matching handles noisy, heavy-tailed data, we change the target distribution from strictly Gaussian to the Student’s t -distribution with varying degrees of freedom [24]. d_f is inversely proportional to the degree of heavy-tailedness, where $d_f = \infty$ corresponds to the Gaussian distribution and $d_f = 1$ corresponds to the infinite variance Cauchy distribution [6]. We plot trends of generalization and memorization under varying heavy-tailed target distributions in Figure 4. First, we briefly note that we do not see any large trends regarding memorization across all degrees of freedom tested. However, we do find consistent patterns for t -distributions with low degrees of freedom, i.e., $d_f \in \{2.0, 3.0\}$ where models reach the memorization threshold earlier in training. Further, in case of infinite variance ($d_f = 1.0$), we see the model both struggles to generalize and begins to memorize a large fraction of the training data earlier in training.

Our results indicate that flow matching reaches a limitation when variance is infinite. We conjecture that under infinite variance (i.e., extreme heavy-tailed data) flow matching struggles to find the consensus direction of the data manifold and compensates by memorizing data instead. To overcome this limitation, we take inspiration from diffusion where the SDE adds Brownian motion to learn trajectories, and this allows the diffusion model to find and adhere to the data manifold before the onset of memorization [25]. Following this idea, we test whether intentionally learning non-linear, curved trajectories stabilizes flow matching training under assumptions of heavy-tailed target distributions.

3.3 Time-Warping for Learning Curved Paths

To induce learning curved paths, we modify the time schedule of the random interpolant against which vector fields are learned by introducing a time schedule [17]. We are motivated by a curvature-based approach due to our observation that, on heavy-tailed data, the final velocity fields in high data density regions point towards the consensus data regions, but in low density regions, velocity fields concentrate all movement towards the extreme outliers (See Figure 5(a) for an example). This phenomenon is also observed in [36], and as the authors have concluded, leads to training instability and affects sample quality. As suggested by [23] to improve DDPMs, we test the use of a time schedule to “time-warp” the loss function and learn curved trajectories. This approach learns non-linear trajectories by changing the scale of time steps based on how far away the random

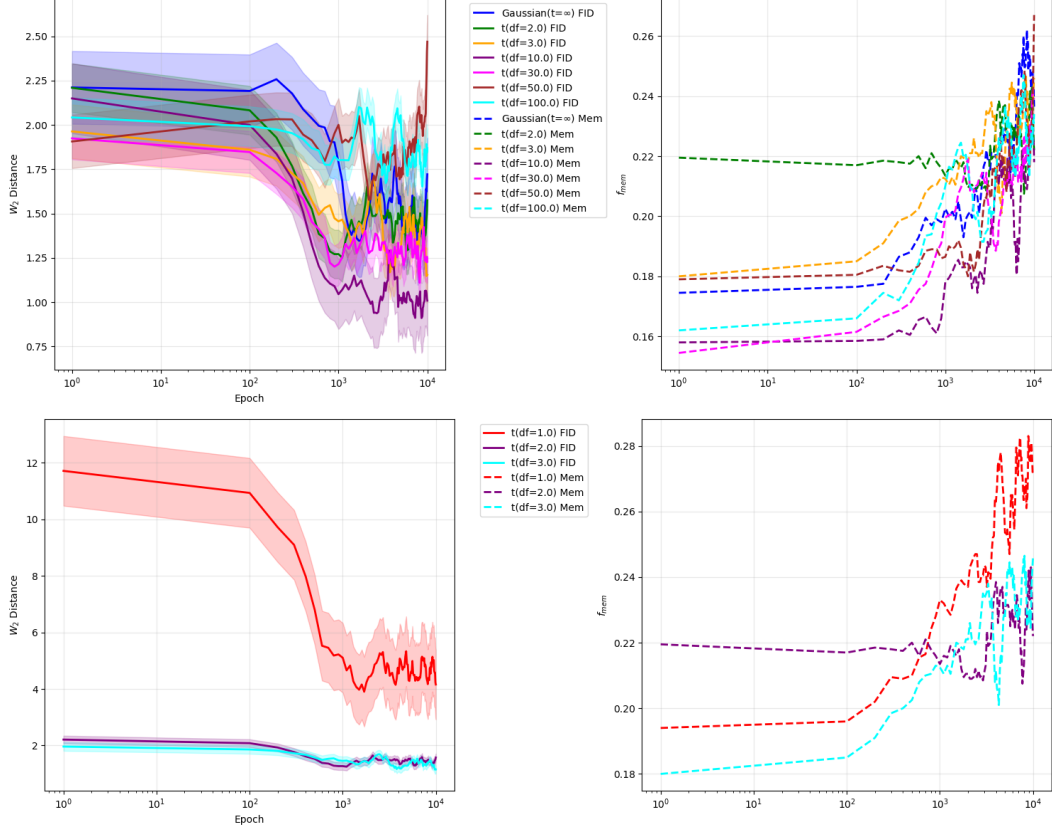


Figure 4: Phase transition for fixed training data size and fixed width flow matching models ($N = 128, W = 512$). Degrees of freedom d_f for Student’s t-distribution take on the values $d_f \in \{1.0, 2.0, 3.0, 10.0, 30.0, 50.0, 100.0, \infty\}$. The top row shows generalization (left) and memorization (right) metrics for all target distribution except for the Cauchy distribution, and the bottom row displays plots explicitly for the Cauchy distribution and two others for comparison.

interpolant is from the training datapoint when calculating the loss. We test the effect of utilizing a power schedule [17]:

$$x_t = (1 - t^k) x_0 + t^k \epsilon, \epsilon \sim \mathcal{N}(0, I)$$

with $k = 2$ and we test a trigonometric schedule [23]:

$$x_t = \left(1 - \sin\left(\frac{\pi}{2}t\right)\right) x_0 + \sin\left(\frac{\pi}{2}t\right) \epsilon, \epsilon \sim \mathcal{N}(0, I)$$

Each schedule warps the vector field in a different manner (See Figure 5(b)). The power schedule learns velocities slowly in regions far from the datapoint on which the flow matching loss is conditioned, and learns velocities quickly in regions close to the data. Conversely, the sine schedule learns velocities quickly in far away areas and gains more resolution closer to the data. The goal is to identify if implementing either schedule changes the generalization-memorization curves for heavy-tailed data, especially for the dataset modeled on the Cauchy distribution.

Our results show that using a sine schedule drastically improves generalization for extreme heavy-tailed data (See Figure 6). Our baseline model shows that flow matching on the Cauchy distribution both generalizes poorly and memorizes quickly once a threshold is met, reaching a Wasserstein distance of around 4.0 at minimum and a f_{mem} of around 0.28 (See Figure 4). Compared with the model trained with a sine schedule, we observe flow matching on the Cauchy distribution now generalizes comparably with the other t-distributions tested. Moreover, we observe that across all distributions, models tended to memorize *less* training data than with the baseline configuration. This suggests that the sine function allows models to both generalize better *and* memorize less data.

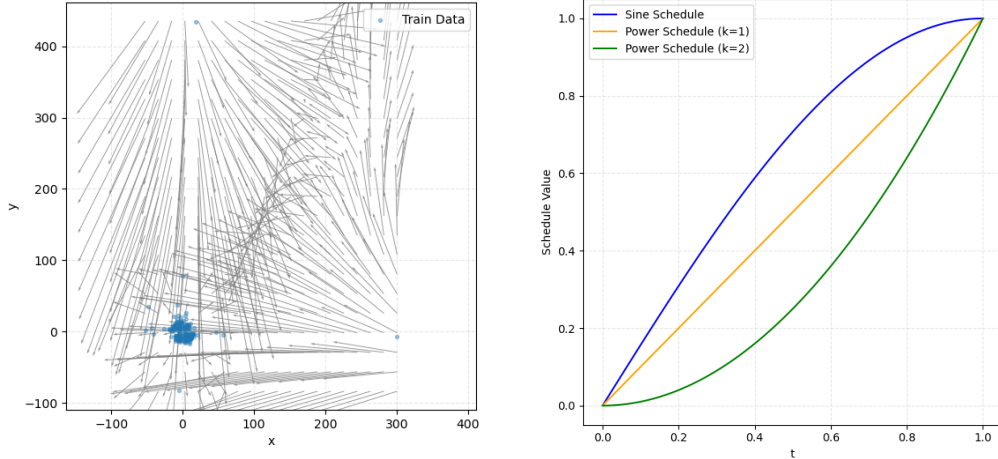


Figure 5: (a) Velocity field at time $t = 1.0$ for a flow matching model trained to match the Cauchy distribution, i.e., Student’s t -distribution with $d_f = 1.0$. Note that in this plot, vectors point in the opposite direction of the actual velocity vectors due to a plotting error. (b) Plots of power and sinusoidal schedules used to warp velocity fields. The power schedule with $k = 2$ and the sine schedule fall on either side of the linear schedule used in standard flow matching implementations.

Further, we observe that the generalization metric diverges across baseline models on heavy-tailed t -distributions with $d_f \geq 2.0$ in the extreme end of the memorization regime (around epoch 10^4), where some models generalize much better than the others (lower Wasserstein distance) and others see a reversal in generalization (Wasserstein distance increases). This divergence in generalization metric is *not* seen in the models trained with a sine schedule, with all models, including the one trained to match the Cauchy distribution, producing stable generalization metrics in the largest epochs. This result suggests that the sine schedule has a stabilizing effect on training across multiple degrees of heavy-tailedness in the target distribution, including for target distributions with infinite variance.

Our experiments with the power schedule do not show many noticeable changes from the baseline, apart from a drastically higher generalization metric for the Cauchy distribution in early epochs. We also observe a similar divergence in generalization metrics for t -distributions with $d_f \geq 2.0$ in later epochs within the memorization regime. These results indicate that learning velocity fields that move points away from data-sparse regions quickly and that move points towards data-rich areas slowly allow flow matching models generalize better under heavy-tailed data assumptions, including for infinite variance distributions. The results also indicate the inverse, which is, that learning velocity fields with greater resolution in data sparse regions and with less resolution in data rich regions either has no change or has a negative effect on generalization.

4 Related Works

There is prior literature on the role of curvature in flow matching. While previous work has argued for learning straight flows [21], [19], [26], this work directly counters the idea that “straighter is better”. Papers have argued for learning curvature as a form of regularization [22], shown that variance in flows is lower in regions close to data than in far away regions [36], and analyzed how and why curvatures arise in flow matching training [18]. Other work on flow matching on Riemannian manifolds have shown that when the data manifold is non-Euclidean, the optimal path is curved in Euclidean space [16], [4]. Further, arguments for explicitly learning curved paths have been made [29]. The “time-warping” method of learning curves using a schedule has been introduced and studied in [1], [23], [17], [24]. However, no recent work has investigated the effect of learning curvature to match heavy-tailed data on the flow matching generalization-memorization phase transition. This is the gap in literature this paper aims to fill.

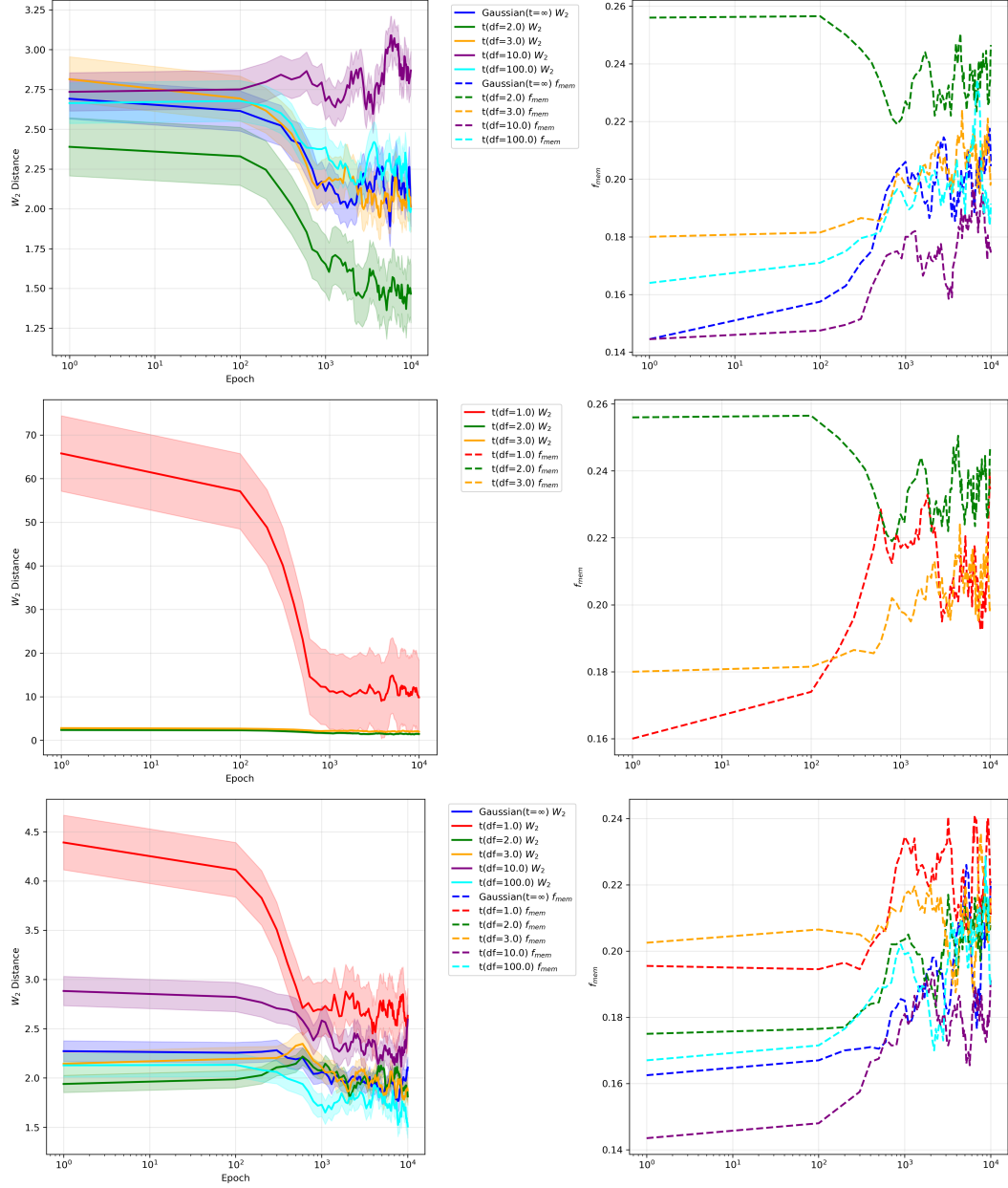


Figure 6: Generalization-memorization phase transitions on varying levels of heavy-tailed data, with fixed training data size and hidden layer width ($N = 128$, $W = 512$), using a power time schedule (top, middle rows) and sine schedule (bottom row). Each row shows the generalization curves (left) and memorization curves (right) over time. Degrees of freedom for target distributions shown are $d_f \in \{1.0, 2.0, 3.0, 10.0, 100.0, \infty\}$, with the exception of the Cauchy distribution in the top row due to scale. The effect of a power schedule on the Cauchy distribution is explicitly show in the middle row, with two other distributions as a point of comparison.

5 Conclusions

This paper studies the generalization to memorization phase transition in flow matching models on 2-dimensional data. Our results find that there is a linear relationship between the amount of training data provided to the model and the onset of memorization, and that an inverse relationship exists between the width of the network and the memorization regime. These results align with the literature on diffusion models applied to realistic image data [3]. This paper also studies the effect of heavy-tailed data on flow matching training. We find that flow matching reaches a limitation under extreme heavy-tailed data, modeled by the infinite variance Cauchy distribution. Furthermore, we identify a novel technique for learning better flow matching models under extreme and very heavy-tailed data by introducing a sinusoidal schedule into the loss function to learn time-warped trajectories. We observe that a sine schedule stabilizes training, helps the model memorize less data, and improve generalization for extreme heavy-tailed data.

Our results challenge the existing literature on learning straight flows [21], [26], [18], as our data shows that learning curved flows may in fact be superior for heavy-tailed data. The idea of modifying the interpolant is grounded in the literature as related work shows that modifying the interpolant indeed makes flow matching more amenable to a wider variety of tasks under fewer assumptions, such as matching non-Euclidean Riemannian data manifolds [16]. We hope the work presented in this paper helps inform alternative conditional flow matching loss functions that allow models to generalize with limited assumptions on the data geometry.

Limitations We note that the choice of Wasserstein-1 metric as a measure of generalization is constrained by the size of the training dataset. While FID (Frechet Inception Distance) is a widely used and agreed upon generalization metric for images, a similar standard does not yet exist for tabular data. As such, generalization metrics are for non-image data often ad-hoc and depend upon saliency to particular datasets [9]. A better metric to measure generalization on the spiral toy dataset studied in this paper could be the Local Intrinsic Dimensionality (LID) since our target distribution naturally lies on a 1-dimensional manifold within 2-dimension space. The use of a metric that is structurally meaningful to the data would help to strengthen the results presented in this paper.

Future Work We suggest the exploration of alternative generalization metrics such as LID for tabular data. We also suggest investigating whether the results from this paper hold up to datasets with dimension > 2 , or to datasets with known heavy-tail to rigorously examine whether explicitly learning curvatures in trajectories via sinusoidal or other non-linear schedules improves flow matching in real-world scenarios. Finally, an examination on the limitations of the sinusoidal schedule may be well advised, to understand the tradeoffs of inducing curvatures in flow matching trajectories.

Notes

While some plots show W_2 on the y-axis, it is a typo and should read W_1 . The Wasserstein-1 metric was used in all relevant calculations.

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A Supplemental Figures

See Figures 7 and 8 for additional plots.

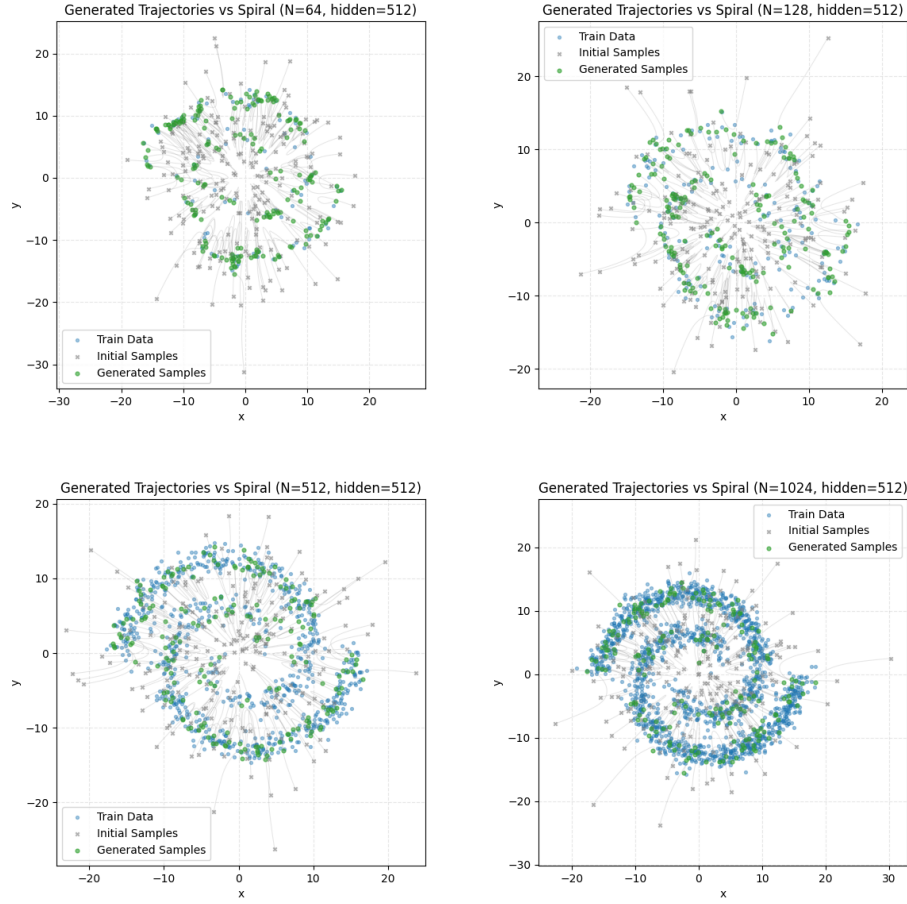


Figure 7: Flow matching trajectories and generated samples superimposed on training data ranging from $N \in \{64, 128, 512, 1024\}$. Signs of memorization is evident in generated samples with $N = 64$, where “beads” of generated data appear close to where the supports of the training data are. This is seen to lesser extent with $N = 128$ and minimally present for $N = 512$ or $N = 1024$. A larger amount of training data improves the ability to learn underlying data manifolds and delays the onset of memorization.

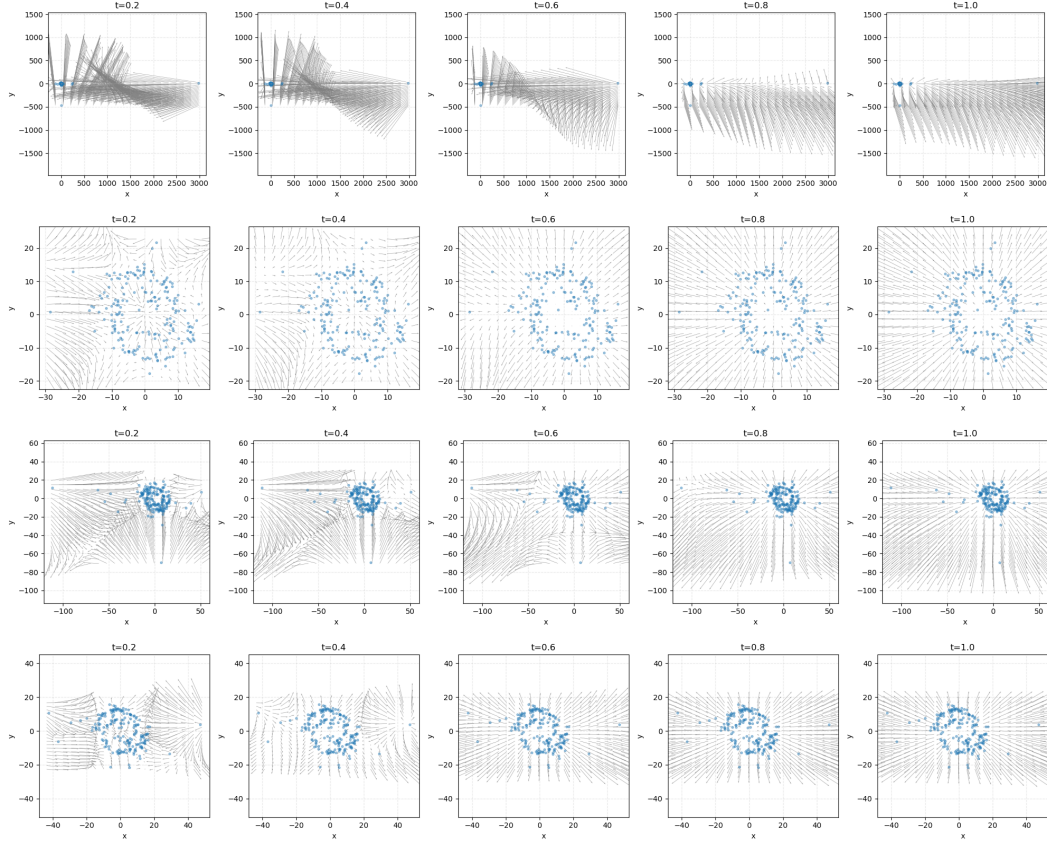


Figure 8: Vector fields for t -distributions using warping time schedules for multiple training time steps. Note that in this plot, vectors point in the opposite direction of the actual velocity vectors due to a plotting error. (Top row) fields from the power schedule on data with $d_f = 1.0$. (second row) fields from the power schedule on data with $d_f = 2.0$. (Third row) fields from the sine schedule on data with $d_f = 1.0$. (Fourth row) fields from the sine schedule on data with $d_f = 2.0$.